

Signal Generators: Slow-Paced Technology



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There are certain instruments that you are bound to find in an electronics lab. A signal generator is one such essential test and measurement (T&M) device; others being a multimeter, power supply, an oscilloscope and so on. Let us go into the details of what is new in this commonly-used instrument.

We have observed that the classification of signal, function and waveform generators is often misunderstood. Let us clarify

that first before moving forward.

Signal generators can be broadly classified into two categories: function generators and arbitrary waveform generators (AWGs).

Function generators. These are used to generate simple, periodic waveforms like sign, square, triangle and sawtooth.

AWGs. These can create complex signals specified by users; these could be periodic or non-periodic and of any possible



National Instruments VirtualBench



Anritsu MG3710A



Keysight UXG Agile signal generators

Bird's-eye view on pricing factors

Demand for signal generators and their cost are mutually-related, says Manish Kwatra, chief executive officer, Metro Electronic Products. Traditionally, these instruments were limited to high-profile research and development (R&D) laboratories. But now we find at least simple function generators in school and college labs, electronics repair centres and so on.

Cost reduces over time. According to Madhukar Tripathi, senior manager - marketing and channel sales, Anritsu India Pvt Ltd, customers in India care less for features and functionality and prefer having devices that cost less. When the price of the device is less, more people start buying new instruments and upgrading existing ones. Kwatra claims that, on balancing out the time value of money, the cost of specific signal generator models from OWEN has reduced by 30 per cent to 40 per cent over a period of two years.

Explaining this process. The price of a standalone signal generator depends on its technical specifications. But due to the continuous evolution of technology and advancements in the T&M arena, value addition for an instrument would be less two years from now. Hence, the cost of that particular model also reduces significantly, explains Avichal Kulshrestha, technical marketing engineer, National Instruments (NI).

Benefit of modular instrumentation. According to Frost & Sullivan's report 'PXI Market to Change the Face of the Test and Measurement Industry,' dated August 19, 2014, open modular platforms such as PXI extensions for instrumentation (PXI) offer high speed of measurement, lower power consumption and flexibility, and reduce redundant components in a test system, thereby bringing down time-to-market and the overall cost of tests.

All-in-one instruments cost less. A notable reduction in cost is happening with the integration of multiple instruments into a single all-in-one device, adds Kulshrestha. Examples are NI VirtualBench and Tektronix MDO3000, which have price tags less than the sum total of the cost of equivalent standalone models.

Offering high-priced models. India is a very price-sensitive market. Customers are sceptical about buying costly signal generators even if these offer new and improved functionalities required by the user. How do manufacturers and distributors face this issue? Tripathi points out that, giving attractive premium rates for newly-launched products would help to an extent.